

# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\hat{L}_{\text{part}} = \mathcal{P}_i \dot{\eta}_i - \frac{1}{2} \mathcal{P}_i^2 + e(\hat{A}_i \dot{X}_i + \hat{A}_0 + \frac{e\theta}{2} \varepsilon_{ij} \hat{A}_i \frac{d}{dt} \hat{A}_j) - \frac{1}{2} \frac{d}{dt} (e\theta \varepsilon_{ij} \mathcal{P}_i \hat{A}_j) .$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\nu^{-1}(\tilde{\varepsilon}) = 4 \tilde{\varepsilon}^{-1} - 8 \frac{n-4}{n-2} \tilde{\varepsilon}^{-2} + 16 \frac{n-4}{n-2} \tilde{\varepsilon}^{-3} - 32 \{ [52 + 108\zeta(3)] - [16 + 36\zeta(3)] n + n^2 \} \frac{\tilde{\varepsilon}^{-4}}{(n-2)^3} + \dots .$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$ds^2 = \frac{1}{4q^2} \left( q \frac{d}{dq} \right)^2 K dq^2 + \left( q \frac{d}{dq} \right) K \left[ \frac{2}{3} d\hat{z}_i d\hat{z}_i - \frac{2}{9} |\hat{z}_i d\hat{z}_i|^2 \right] + \frac{4}{9} \left( q^2 \frac{d^2}{dq^2} \right) K |\hat{z}_i d\hat{z}_i|^2 .$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\frac{4}{\tilde{R}} h^{-1/4} \pi_i h^{1/2} h^{ij} \pi_j h^{-1/4} = -\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \left( i \frac{\partial}{\partial \varphi} + \frac{\sigma^3}{2} \cos \theta \right)^2 .$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\lim_{\theta \rightarrow 1} \left( \Omega, S_{\theta(\mathcal{L}_0 + \epsilon \mathcal{L}_1)}(\lambda f) \Omega \right) = \lim_{\theta \rightarrow 1} \frac{\left( \Omega, S_{\theta \mathcal{L}_0}(\theta \epsilon \mathcal{L}_1 + \lambda f) \Omega \right)}{\left( \Omega, S_{\theta \mathcal{L}_0}(\theta \epsilon \mathcal{L}_1) \Omega \right)}$$